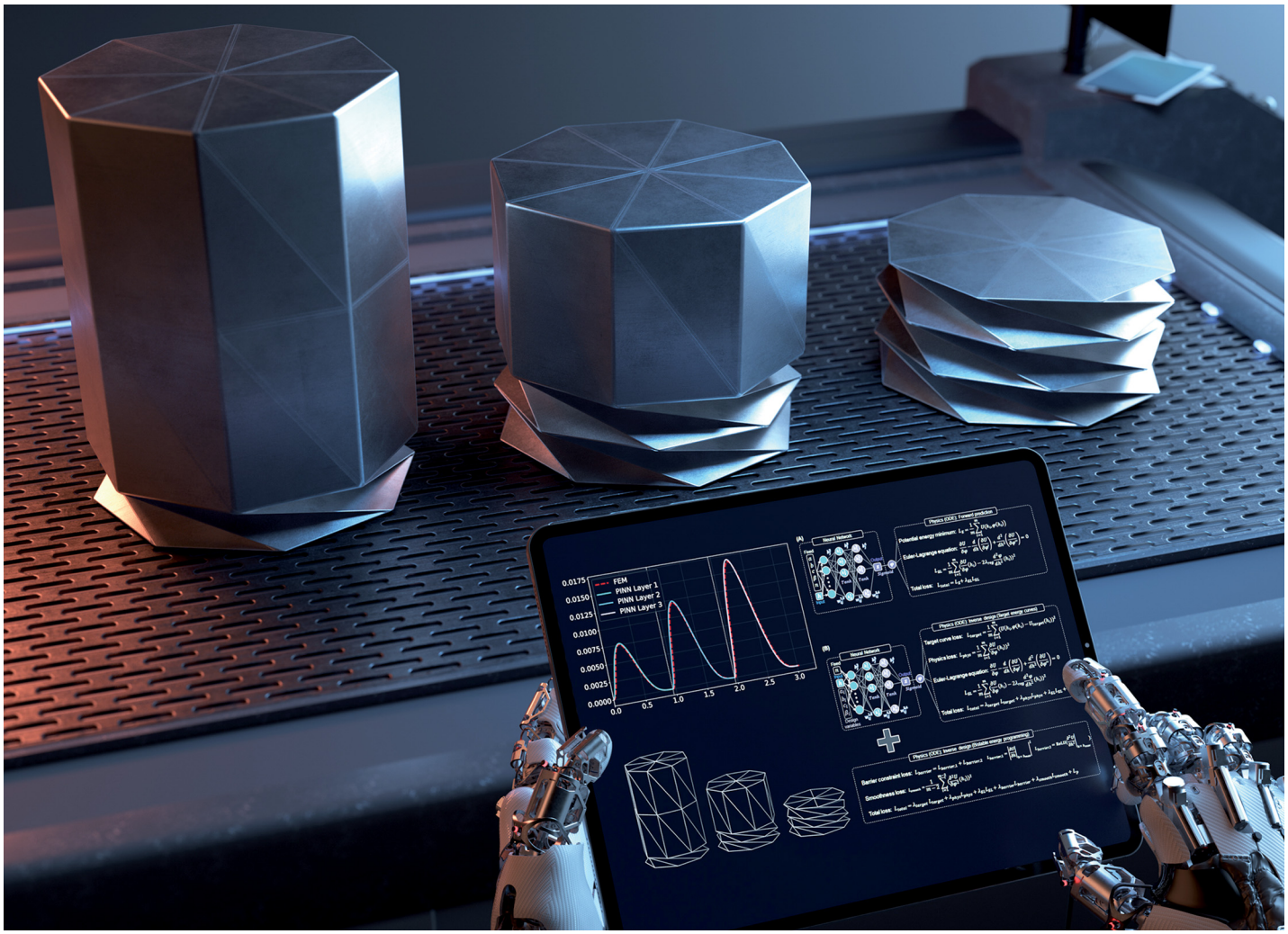


Showcasing research from Professor Seunghwa Ryu's group at the Department of Mechanical Engineering and the InnoCORE PRISM-AI Center, KAIST, Daejeon, Republic of Korea

Physics-informed neural networks for programmable origami metamaterials with controlled deployment

A physics-informed neural network framework is developed to predict and design full energy landscapes of conical Kresling origami without requiring training data. By embedding mechanical equilibrium into the learning process, the method enables precise control of stable states and energy barriers and supports bistable programming from minimal inputs. Extending this approach to multilayer assemblies further allows programmable and experimentally validated sequential deployment.

Image reproduced by permission of Seunghwa Ryu from *Mater. Horiz.*, 2025, **12**, 10641.

As featured in:



See Jinkyu Yang, Seunghwa Ryu *et al.*, *Mater. Horiz.*, 2025, **12**, 10641.

ROYAL SOCIETY OF CHEMISTRY

rsc.li/materials-horizons

Registered charity number: 207890

Materials Horizons

ROYAL SOCIETY OF CHEMISTRY

COMMUNICATION

View Article Online
View Journal | View Issue

Published on 07 November 2025. Downloaded by University of Minnesota - Twin Cities on 5/18/2026 4:06:03 AM.

Check for updates

Cite this: *Mater. Horiz.*, 2025, **12**, 10641

Received 21st August 2025,
Accepted 5th November 2025

DOI: 10.1039/d5mh01607j

rsc.li/materials-horizons

Physics-informed neural networks for programmable origami metamaterials with controlled deployment

Sukheon Kang,[a] Youngkwon Kim,[b] Jinkyu Yang*[b] and Seunghwa Ryu *[ac]

Origami-inspired structures provide unprecedented opportunities for creating lightweight, deployable systems with programmable mechanical responses. However, their design remains challenging due to complex nonlinear mechanics, multistability, and the need for precise control of deployment forces. Here, we present a physics-informed neural network (PINN) framework for both forward prediction and inverse design of conical Kresling origami (CKO) without requiring pre-collected training data. By embedding mechanical equilibrium equations directly into the learning process, the model predicts complete energy landscapes with high accuracy while minimizing non-physical artifacts. The inverse design routine specifies both target stable-state heights and separating energy barriers, enabling freeform programming of the entire energy curve. This capability is extended to hierarchical CKO assemblies, where sequential layer-by-layer deployment is achieved through programmed barrier magnitudes. Finite element simulations and experiments on physical prototypes validate the designed deployment sequences and barrier ratios, confirming the robustness of the approach. This work establishes a versatile, data-free route for programming complex mechanical energy landscapes in origami-inspired metamaterials, offering broad potential for deployable aerospace systems, morphing structures, and soft robotic actuators.

New concepts

We introduce a mechanics-first learning paradigm that transforms origami geometry into programmable energy landscapes. Our physics-informed neural network (PINN) embeds the truss-based equilibrium of conical Kresling origami (CKO) directly into the loss function, eliminating the need for precomputed datasets and suppressing non-physical artifacts. While prior PINN and optimization methods attempt to match entire energy curves or discrete data points, our approach formulates an inverse-design scheme that relies on only minimal specifications: two stable heights and a target energy barrier. From these inputs, the model autonomously infers the barrier location and reconstructs a physically admissible energy profile. The same principle extends to hierarchical assemblies, where prescribing layer-wise barrier magnitudes enables passive, sequential deployment without feedback or complex control systems. Conceptually, this framework reframes the design of origami structures, and more broadly the design of multistable metamaterials, as a problem of energy-landscape programming under physics constraints. It provides an interpretable and differentiable approach that generalizes across varying conical geometries and polygon counts. This paradigm shift opens a pathway for rapid, fabrication-aware exploration of structures intended for deployable aerospace systems, morphing skins, and soft robotic actuators.

1. Introduction

Origami-inspired mechanical systems offer a unique path to lightweight yet multifunctional structures, combining high reconfigurability with geometric design freedom.[1,2] This combination enables motion profiles, deployment behaviors, and adaptive functions that conventional engineering designs cannot easily achieve, opening new opportunities across aerospace,[3,4] robotics,[5–7] energy devices,[8–11] and mechanical metamaterials.[12,13] Among the various origami motifs, Kresling origami originates from the periodic wrinkle patterns formed during torsional buckling of thin cylindrical shells and exhibits a distinctive kinematic coupling between axial contraction and twisting during folding.[14] By tuning only a few geometric parameters, Kresling origami can exhibit multistability, with its energy landscape programmable purely through geometry.[15–18] These properties have led to its use in deployable structures, soft robotic actuators, impact-mitigation devices, reconfigurable antennas, and other advanced systems.[19–25]

Understanding and exploiting these unique mechanical responses requires accurate modeling approaches and efficient design strategies. To model and design Kresling origami, most previous studies have used truss-based or continuum

[a] *Department of Mechanical Engineering, Korea Advanced Institute of Science and Technology (KAIST), Daejeon 34141, Republic of Korea. E-mail: ryush@kaist.ac.kr*
[b] *Department of Mechanical Engineering, Seoul National University, Seoul 08826, Republic of Korea. E-mail: jkyang11@snu.ac.kr*
[c] *KAIST InnoCORE PRISM-AI Center, Korea Advanced Institute of Science and Technology (KAIST), Daejeon 34141, Republic of Korea*

Published on 07 November 2025. Downloaded by University of Minnesota - Twin Cities on 5/18/2026 4:06:03 AM.

mechanics approaches combined with conventional optimization algorithms.[26–29] While effective for fixed geometries or targeting specific stable states, these methods face significant challenges in addressing high-dimensional objectives, such as tailoring nonlinear energy landscapes, achieving multistability, or programming precise deployment sequences, and they often lose generality when applied to more complex architectures.

Recent advances in machine learning offer complementary pathways to overcome these modeling limitations. Data-driven approaches have emerged, training deep learning models on large finite element datasets to predict or inversely design the mechanical behavior of architected metamaterials,[30–33] including origami structures.[34–37] However, these approaches require extensive labeled datasets and often fail to generalize beyond the predefined design space. Physics-informed neural network (PINN) approaches address this limitation by embedding governing equations directly into the learning process,[38] enabling design without labeled data. PINNs have been applied to topology optimization, metamaterial design, and energy-based structural tailoring.[39–43]

Within this growing field of research, early efforts have begun to explore PINN-based origami design. For Kresling origami, Liu *et al.*[44] made an important first step by applying a PINN-based approach to predict the energy curve of cylindrical configurations and demonstrate a basic inverse design. Their framework, designed for axially symmetric geometries, focused on matching discrete points on the energy curve and did not aim to control barrier heights or shape the complete energy landscape required for prescribed deployment paths. Finite element analysis (FEA) and experimental validation were beyond the scope of that study. Furthermore, the concept of assembling multilayer structures and programming energy barriers in a hierarchical manner to achieve sequential deployment remains largely unexplored, despite its importance for deployable antennas, morphing aerospace skins, and continuously actuated soft robotic systems.

Beyond origami mechanisms, shell- and plate-based programmable metamaterials have also demonstrated multistable behaviors and tunable energy landscapes by leveraging geometric nonlinearity and snapping elements. Representative examples include morphing metastructures, popper-like spherical 0shells, dome-patterned sheets, and shallow-arch chains.[45–48] These studies broaden the context of programmable metamaterials, highlighting that geometry-induced multistability and tunable energy landscapes can be achieved across diverse structural forms.

Building on these foundations, the present study introduces an integrated PINN framework for both forward prediction and inverse design of conical Kresling origami (CKO). The framework predicts full energy landscapes without pre-collected training data and directly maps geometric parameters to mechanical responses. Enhanced static-equilibrium constraints allow generalization beyond cylindrical to arbitrary conical geometries while preventing non-physical artifacts. The inverse design routine specifies both the target heights of stable states and the magnitudes of the separating energy barriers, enabling freeform control over the entire energy curve. Leveraging this capability, we design hierarchical CKO assemblies with programmed barriers that achieve controlled, layer-by-layer deployment. The approach is validated through FEA and physical prototypes, demonstrating robust agreement in deployment order and barrier ratios.

2. Methodology

2.1. Geometric, kinematic, and energy modeling of conical Kresling origami

The CKO structure consists of periodically arranged quadrilateral unit cells folded from a flat crease pattern (Fig. 1(A) and (B)). Each unit cell is defined by five geometric parameters: the top and bottom edge lengths a and b, the mountain and valley crease lengths c and d, and the internal angle β between the bottom edge and the adjacent mountain crease. By folding the mountain and valley creases of n unit cells and connecting them along the lateral edges, the structure forms a conical shell in which the top and bottom polygonal cross-sections differ in radius.[49,50] This configuration induces a characteristic deformation mode in which axial contraction is coupled with rotational twisting.

The deformation of the structure is fully described by two state variables: the vertical height h between the polygons and the in-plane rotation angle φ of the top polygon relative to the bottom. We define $\varphi = 0$ as the reference configuration in which the side triangles are symmetrically aligned without in-plane twist, and the structure exhibits no chirality. In the present elastic truss-based model, each crease is modeled as a linear axial spring, allowing small extensional and compressional deformation along the crease direction. Consequently, h and φ are energetically coupled through the total potential energy $U(h, \varphi)$. Under displacement-controlled loading, the equilibrium rotation $\varphi(h)$ is determined by the torque-free condition:

$$\frac{\partial U}{\partial \varphi}(h, \varphi(h)) = 0 \tag{1}$$

which defines the physically realizable states along an effective one-degree-of-freedom equilibrium path (see SI Section S1). These energetically defined equilibrium states form the physical basis for the learning framework described in the next section. As shown in Fig. 1(C), compression leads to simultaneous reduction in height and increase in rotation, producing smooth and continuous folding motion. For clarity, both physical paper models and their truss abstractions are shown at four representative deformation stages.

To model the mechanical response, we adopt a truss-based abstraction in which each mountain and valley crease is treated as a linear axial spring resisting extension and compression. We note in passing that the CKO is a non-rigid origami, experiencing evident facet deformation during folding motions. Thus, this axial spring model not only represents the crease contraction/extension, but also encompasses more complicated facet and crease deformation (*e.g.*, bending and twisting motions

 This journal is © The Royal Society of Chemistry 2025

Published on 07 November 2025. Downloaded by University of Minnesota - Twin Cities on 5/18/2026 4:06:03 AM.

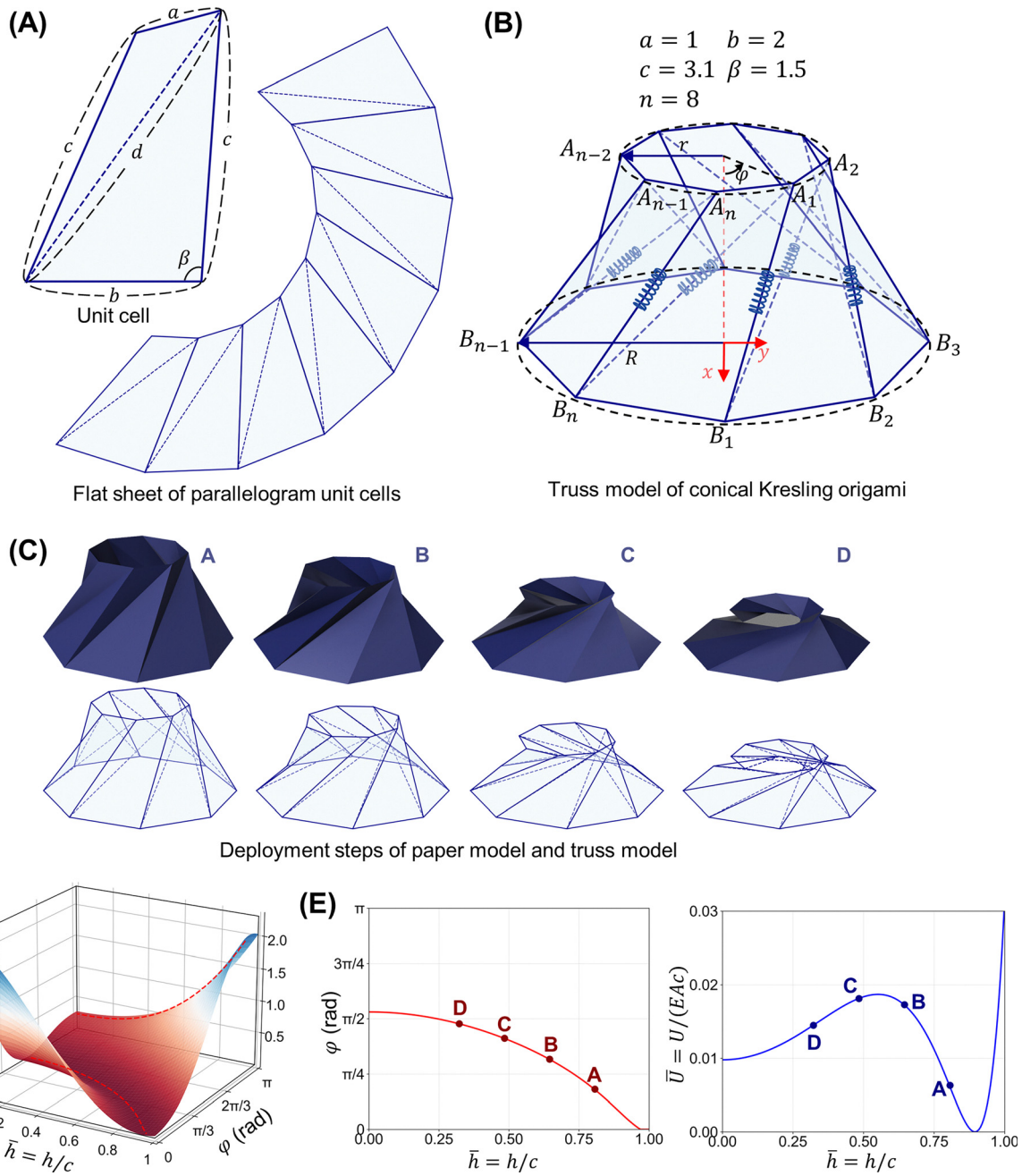


Fig. 1 Geometric modeling and energy landscape of conical Kresling origami (CKO). (A) Unit cell and flat crease pattern with geometric parameters: *a*, *b*, *c*, *d*, and β. (B) 3D CKO structure and truss model showing vertices (A_i,B_i), radii (*r*, *R*), and rotation angle φ (a = 1, b = 2, c = 3.1, β = 1.5, n = 8). (C) Compression sequence (A)–(D) with paper model (top) and truss model (bottom). (D) Energy landscape $U(\bar{h},\varphi)$ with torque-free equilibrium paths (red dashed lines). (E) Energy U and rotation angle φ *versus* normalized height $\bar{h}$, with points A–D corresponding to stages in (C).

of facets, and axial, shear and torsional motions of creases). Under this assumption, the total strain energy of the structure is computed as the sum of elastic energies stored in the creases:

$$U(h,\varphi) = \frac{n}{2}k_{\mathrm{m}}\left(L_{\mathrm{m}} - L_{\mathrm{m}}^{0}\right)^{2} + \frac{n}{2}k_{\mathrm{v}}\left(L_{\mathrm{v}} - L_{\mathrm{v}}^{0}\right)^{2} \qquad (2)$$

Here, L_{m} and L_{v} denote the current lengths of the mountain and valley creases, and L_{m}^{0}, L_{v}^{0} are their rest lengths in the unfolded state. The stiffness values $k_{\mathrm{m}} = EA_{\mathrm{m}}/L_{\mathrm{m}}^{0}$ and $k_{\mathrm{v}} = EA_{\mathrm{v}}/L_{\mathrm{v}}^{0}$ are defined by the Young's modulus E and the cross-sectional areas A_{m} and A_{v} of the axial truss members, which are normalized to unity in this work. Because the total strain energy is linearly proportional to EA, all quantities are expressed in a dimensionless form normalized by EA to focus on the shape of the energy landscape rather than its absolute magnitude. This normalization ensures that E and A act only as global scaling factors without affecting the equilibrium path or bistable characteristics. The same normalization is consistently applied in

Published on 07 November 2025. Downloaded by University of Minnesota - Twin Cities on 5/18/2026 4:06:03 AM.

both the theoretical formulation and the FEA comparisons. In this formulation, each crease is represented by an axial spring, and explicit rotational (hinge) and buckling terms are intentionally omitted. This abstraction captures the primary geometric nonlinearity governing the multistable behavior while keeping the energy formulation simple and well suited for physics-informed learning.

The resulting energy landscape $U(h, \varphi)$ is a two-dimensional surface composed of valleys and ridges that capture the multistable nature of the structure. As shown in Fig. 1(D), the red dashed curves trace equilibrium paths where the torque vanishes, *i.e.*, $\partial U/\partial\varphi = 0$. These represent the physically realizable rotation states at each height. Fig. 1(E) presents the energy and rotation profiles along the minimum-energy branch, where the system follows the lowest-energy trajectory under axial loading. The four deformation stages in Fig. 1(C) are marked for reference. This framework captures essential CKO mechanics and forms the physical basis for the learning models described in the next section. Full derivations are provided in SI Section S1.

2.2. Physics-informed neural network framework

2.2.1 Forward prediction using PINN. In the forward design problem, the goal is to predict the potential energy landscape of a CKO with given geometry by learning the deformation-dependent rotation $\varphi(h)$ using a PINN. The neural network architecture, shown in Fig. 2(A), is a fully connected feedforward model with two hidden layers of 128 tanh-activated neurons. The input layer receives the height h, while geometric parameters (a, b, c, β, n) are treated as fixed constants. To ensure physical feasibility, the rotation angle φ is constrained to $[0, \varphi_{\text{upper}}]$ through a sigmoid-based scaling:

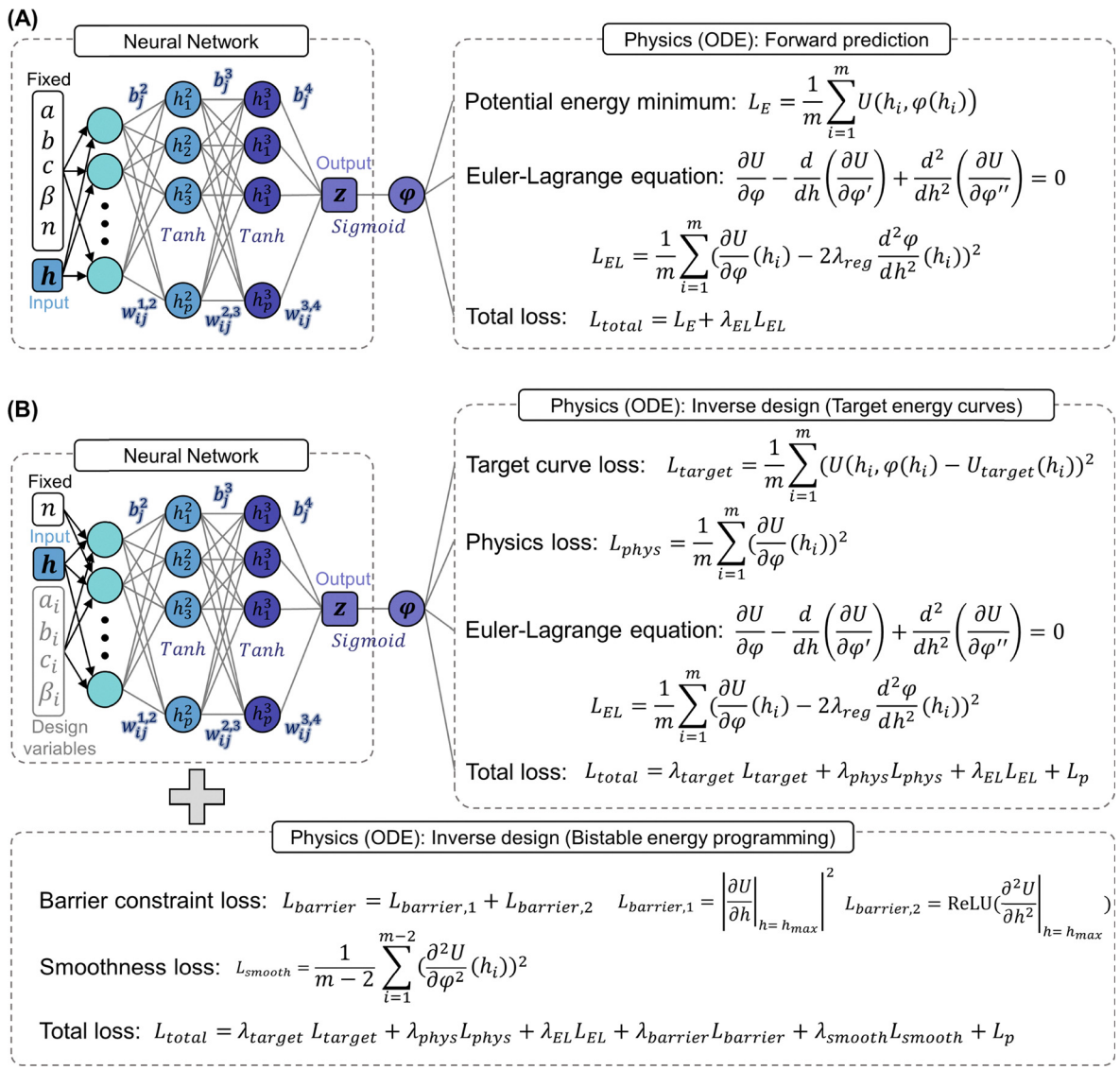


Fig. 2 Physics-informed neural network (PINN) framework. (A) Forward prediction: network predicts $\varphi(h)$ from height h with energy minimization (L_E) and Euler–Lagrange regularization (L_{EL}). (B) Inverse design: joint optimization of structural parameters (a, b, c, β) and $\varphi(h)$. Base loss includes L_{target}, L_{phys}, L_{EL}, and $L_{penalty}$. For bistable programming, additional $L_{barrier}$ and L_{smooth} terms enable minimal specification design.

 This journal is © The Royal Society of Chemistry 2025

Published on 07 November 2025. Downloaded by University of Minnesota - Twin Cities on 5/18/2026 4:06:03 AM.

$$\varphi(h) = \varphi_{\text{lower}} + (\varphi_{\text{upper}} - \varphi_{\text{lower}})\cdot\sigma(z(h)) \tag{3}$$

where $z(h)$ is the raw network output, $\sigma(\cdot)$ is the sigmoid function, and $\varphi_{\text{upper}} = \min(\varphi_{\text{f}}, \varphi_{\text{s}})$ with φ_{f} being the folded-flat angle and $\varphi_{\text{s}} = \pi - 2\pi/n$ the locked configuration.

The network is optimized using a variational objective functional based on minimum total potential energy:

$$J[\varphi] = \int_{h_{\min}}^{h_{\max}} U(h, \varphi(h))\mathrm{d}h \tag{4}$$

With an Euler–Lagrange (EL) regularization term:

$$\frac{\partial U}{\partial \varphi} - 2\lambda_{\text{reg}}\frac{\mathrm{d}^2\varphi}{\mathrm{d}h^2} = 0 \tag{5}$$

The total loss function combines:

$$L_{\text{total}} = L_{\text{E}} + \lambda_{\text{EL}}L_{\text{EL}} \tag{6}$$

$$L_{\text{E}} = \frac{1}{m}\sum_{i=1}^{m} U(h_i, \varphi(h_i)) \tag{7}$$

$$L_{\text{EL}} = \frac{1}{m}\sum_{i=1}^{m}\left(\frac{\partial U}{\partial \varphi}(h_i) - 2\lambda_{\text{reg}}\frac{\mathrm{d}^2\varphi}{\mathrm{d}h^2}(h_i)\right)^2 \tag{8}$$

The energy loss L_{E} minimizes total potential energy, ensuring physical consistency, while the Euler–Lagrange regularization L_{EL} acts as a variational constraint that suppresses nonphysical kinks or spurious extrema in multistable energy landscapes.

In addition, the torque-free constrain ($\partial U/\partial\varphi = 0$) enforces rotational equilibrium so that the learned trajectory remains on physically realizable equilibrium states throughout deformation. A complete description of all loss components, including the smoothness and barrier terms used in subsequent sections, is provided in SI Section S2.2. The detailed training procedure is provided in Algorithm 1 (SI Section S2).

2.2.2 Inverse design based on target energy curve. In inverse design, both the structural parameters (a, b, c, β) and $\varphi(h)$ are optimized to match a target energy profile. The network architecture remains identical to the forward case, but structural parameters are now encoded as differentiable variables constrained *via* logarithmic and sigmoid transformations (Fig. 2(B)).

The total loss function is:

$$L_{\text{total}} = \lambda_{\text{target}}L_{\text{target}} + \lambda_{\text{phys}}L_{\text{phys}} + \lambda_{\text{EL}}L_{\text{EL}} + L_{\text{penalty}} \tag{9}$$

$$L_{\text{target}} = \frac{1}{m}\sum_{i=1}^{m}\left(U(h_i, \varphi(h_i)) - U_{\text{target}}(h_i)\right)^2 \tag{10}$$

$$L_{\text{phys}} = \frac{1}{m}\sum_{i=1}^{m}\left(\frac{\partial U}{\partial \varphi}(h_i)\right)^2 \tag{11}$$

$$L_{\text{penalty}} = \begin{cases} 0, & \text{physical constraints are satisfied} \\ C, & \text{otherwise} \end{cases} \tag{12}$$

with $C \gg 1$ (*e.g.*, $C = 10^3$).

Here, L_{target} enforces energy curve matching, L_{phys} ensures torque-free equilibrium ($\partial U/\partial\varphi = 0$), and L_{EL} (as defined in eqn (8)) maintains variational consistency. The penalty term L_{penalty} enforces geometric feasibility:

$$\begin{aligned} &|r - R| \le c \le r + R \,(\text{Mountain crease feasibility}) \\ &d > 0 \,(\text{Valley crease positivity}) \\ &\beta \in (0, \pi) \,(\text{Angle validity}) \\ &-1 \le \cos(\varphi_{\text{f}}) \le 1 \,(\text{Flat foldability condition}) \end{aligned} \tag{13}$$

The training procedure (Algorithm 2, SI Section S2) iteratively updates both parameters and network weights *via* gradient descent.

2.2.3 Inverse design based on bistable energy programming. This framework extends inverse design to underdetermined problems requiring only minimal specifications: two stable heights (h_1, h_2) with zero energy and a target barrier value U_{barrier}. The barrier location is autonomously inferred by the model.

The total loss function incorporates additional terms:

$$\begin{aligned} L_{\text{total}} = \lambda_{\text{target}}L_{\text{target}} + \lambda_{\text{phys}}L_{\text{phys}} + \lambda_{\text{EL}}L_{\text{EL}} + \lambda_{\text{barrier}}L_{\text{barrier}} \\ + \lambda_{\text{smooth}}L_{\text{smooth}} + L_{\text{penalty}} \end{aligned} \tag{14}$$

$$L_{\text{target}} = \frac{1}{k}\sum_{i=1}^{k}\left(U(h_i, \varphi(h_i)) - U_{\text{target}}(h_i)\right)^2 \tag{15}$$

where L_{phys} (eqn (11)) and L_{EL} (eqn (8)) remain as previously defined. The novel barrier constraint loss ensures proper extremum identification:

$$L_{\text{barrier}} = L_{\text{barrier},1} + L_{\text{barrier},2} \tag{16}$$

$$L_{\text{barrier},1} = \left|\frac{\partial U}{\partial h}\Big|_{h=h_{\max}}\right|^2 \tag{17}$$

$$L_{\text{barrier},2} = \text{Re}LU\left(\frac{\partial^2 U}{\partial h^2}\Big|_{h=h_{\max}}\right) \tag{18}$$

$$L_{\text{smooth}} = \frac{1}{m-2}\sum_{i=1}^{m-2}\left(\frac{\partial^2 U}{\partial \varphi^2}(h_i)\right)^2 \tag{19}$$

The barrier location is defined as $h_{\max} = \text{argmax}\{U(h)|h \in [h_1, h_2]\}$. At this point, $L_{\text{barrier},1}$ enforces zero gradient at the barrier, while $L_{\text{barrier},2}$ ensures convexity *via* ReLU activation. L_{smooth} serves as a global regularization term that maintains continuity and smoothness of the learned energy curve across the entire deformation path, preventing abrupt curvature transitions and enhancing the numerical stability of multistable energy landscapes. The constraint penalty follows eqn (12) with conditions from eqn (13). Fig. 2(B) illustrates the complete architecture, with training details in Algorithm 3 (SI Section S2).

2.3 Experimental methods

Three zigzag type CKO prototypes (layer 1, 2, and 3) were fabricated using 0.2 mm-thick polyethylene terephthalate (PET) sheet. The top and bottom polygonal plates were made of 2 mm-thick acrylic plates. All PET and acrylic were cut by a

Published on 07 November 2025. Downloaded by University of Minnesota - Twin Cities on 5/18/2026 4:06:03 AM.

laser cutting machine (Trotec Speedy 360). PET sheets were connected to the acrylic plates with M3 bolts and nuts, and the two ends of each PET panel joints were joined using double-sided foam tape.

The geometric configuration of the prototypes was a zigzag CKO with a fixed polygonal order of n = 8 (regular octagon). Edge lengths a = 24.5 mm and b = 49 mm were constant, while the mountain crease length c and the internal angle β varied as shown in Fig. 3(A): layer 1 (c = 83.1 mm, β = 87.6°), layer 2 (c = 85.9 mm, β = 85.5°), layer 3 (c = 87.8 mm, β = 83.8°).

A multi-layer CKO assembly was formed by stacking Layer 1, Layer 2, and Layer 3 coaxially between four acrylic plates: a base plate (Layer 1 bottom), two interface plates – IP12 (shared between Layer 1 top and Layer 2 bottom) and IP23 (shared between Layer 2 top and Layer 3 bottom), and a top cap plate (Layer 3 top). To achieve one-to-one registration of vertices and bolt-hole patterns at the interface plates, Layer 2 was placed upside down, there by exchanging its wider bottom and narrower top panels. Each layer was connected to the plates with M3 bolts and nuts.

To improve foldability and durability, a double-stitch perforation (DSP) pattern was applied based on the design methodology proposed in previous studies.[51] The DSP pattern permits shear and twist along the crease, reduces fatigue and fracture, and is well- suited for non-rigid origami mechanisms such as the CKO. The DSP parameters (L, B, H, W) were based on the bi-stable configuration of Kresling structures. Parameters (L, B, H, W) = (10, 1.0, 0.4, 0.6) mm for the a and b creases, and (10, 1.0, 0.4, 0.4) mm were for the c and d creases as shown in Fig. 3(A).

Experimental tests were conducted to validate the theoretical model and examine the mechanical behavior of the CKO. The experimental setup, shown in Fig. 3(B), consisted of a CKO mounted on an in-house load frame with displacement controlled by a linear stage (Velmex Inc., Bislide MN10). Force was measured using a tension/compression load cell (LUX-B-200N-ID), with a sleeve bearing placed below the load cell to allow free rotation of the top plate and enable the natural twisting motion of the CKO. Compression was applied slowly under quasi-static conditions so that rate-dependent effects were negligible.

Two specimen types were tested: a single-layer CKO and a multi-layer CKO assembly. For each type, uniaxial compression tests were performed three times, and the averaged force data were used for analysis. Force–displacement curves were obtained directly from the measured data, and energy–displacement curves were calculated by numerical integration of the force over displacement.

2.4 Finite element analysis

To validate the predicted energy responses of CKO structures, FEA was conducted using the ABAQUS/Implicit 2017 solver from Dassault Systèmes. The simulation employed a beam-truss model consistent with the truss-based formulation described in Mountain and valley creases were modeled as linear elastic T3D2 (2-node linear 3-D truss) elements with Young's modulus E = 1 MPa, cross-sectional area A = 20 mm^2, and Poisson's ratio 0.3.[49] Self-contact interactions and frictional effects were not considered to isolate elastic truss deformation. To simulate rigid behavior, the top and bottom polygonal edges were assigned a modulus 100 times greater than that of the crease elements. The geometric dimensions were uniformly scaled by a factor of 100 relative to the nominal dimensions used in the PINN framework. This scaling was applied solely to improve the numerical conditioning of the implicit solver and to keep displacement, force, and strain–energy magnitudes within recommended ranges. All FEA energy values were subsequently normalized by the same EA factor used in the theoretical model so that comparisons with the PINN predictions were performed on a common dimensionless basis. The nominal constants E = 1 MPa, cross-sectional area A = 20 mm^2 therefore serve only as scaling parameters and do not influence the energy landscape shape or equilibrium trajectory.

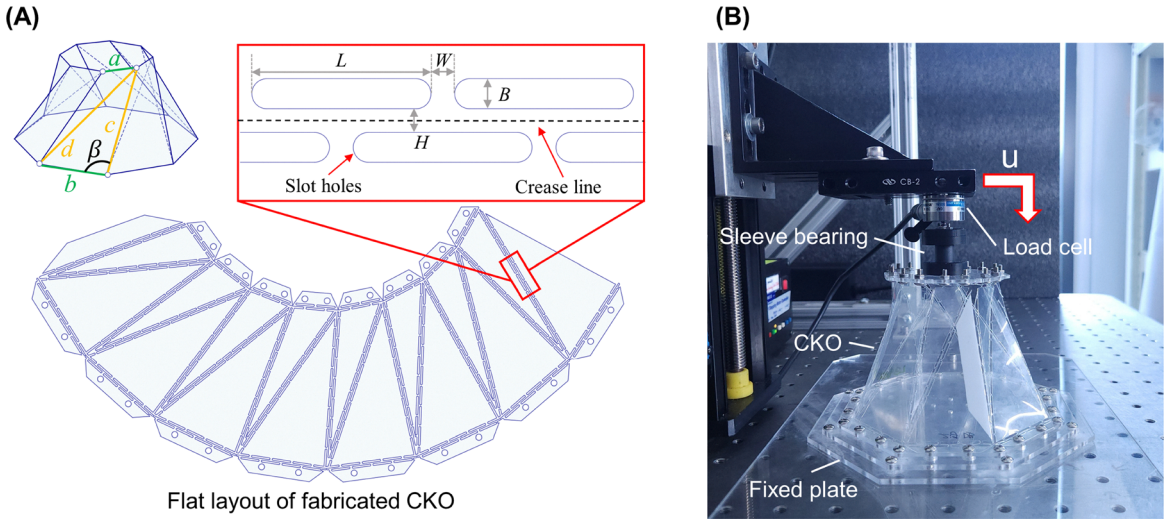


Fig. 3 Geometry and experimental setup of a CKO. (A) Geometric design and crease parameters of the CKO unit. (B) Experimental setup for uniaxial compression test of the CKO.

 This journal is © The Royal Society of Chemistry 2025

Published on 07 November 2025. Downloaded by University of Minnesota - Twin Cities on 5/18/2026 4:06:03 AM.

Two types of simulations were performed depending on the configuration. In the single-layer CKO configuration, energy-*versus*-height profiles $U(h)$ were obtained by first compressing the structure from its zero-energy equilibrium height to a fully folded state, and then gradually stretching it while recording the energy over a range of increasing heights. During this process, the bottom polygon was fixed, and displacement control was applied to the top polygon.

For multi-layer CKO assemblies, the focus was placed on measuring energy as a function of height change rather than absolute height. Each structure was compressed from its equilibrium state, and the total strain energy was computed over the deformation process. This approach captures the energy evolution associated with sequential deployment under external loading.

The FEA results were quantitatively compared with those obtained from the PINN-based models introduced in Sections 2.2.1–2.2.3, including forward predictions, inverse designs, and energy barrier programming in multi-unit assemblies.

3. Results

3.1 Forward prediction of energy curve

We first evaluate the forward prediction capability of the proposed PINN framework by testing its accuracy in predicting the deformation-dependent energy response $\bar{U}(\bar{h})$ for CKO structures with diverse geometries. Fig. 4(A) shows the energy landscape computed from the truss-based model over the normalized design space ($\bar{h}$, φ), with torque-free equilibrium paths ($\partial U/\partial\varphi = 0$). The PINN-predicted paths (blue dashed) coincide with these equilibrium curves across all cases, confirming that the model identifies physically valid solutions solely from embedded mechanical constraints without labeled data.

To examine the predicted energy curves in more detail, Fig. 4(B) compares the PINN predictions (blue solid lines) with FEA results (black dots) for four representative geometries chosen to illustrate distinct energy landscape features.

- Case 1: classical bistability with two zero-energy minima.
- Case 2: near-monostability with a single dominant zero-energy minimum.
- Case 3: asymmetric bistability, with one zero-energy minimum and another at a non-zero energy level.
- Case 4: presents a zero-energy stable state near $\bar{h} = 0$ (fully folded configuration) and another at higher $\bar{h}$, serving as a critical structure for the inverse design and sequential deployment demonstrations.

In Fig. 4(B), red markers identify the zero-energy stable points, with corresponding 3D renderings shown in red. Green markers indicate local maxima (energy barriers) whose 3D configurations are shown in green. These annotations highlight the diversity and programmability of the CKO energy landscape, from symmetric bistability to highly asymmetric responses.

In all cases, the predicted curves closely follow the FEA results, capturing both the shape of the energy profile and the positions of stable and unstable states. Minor discrepancies are negligible and do not affect the qualitative features of the energy landscape. Quantitatively, Table 1 reports the root mean square error (RMSE) and coefficient of determination (R^2) between PINN and FEA results, with all four cases achieving $R^2 > 0.93$ and RMSE values within 5×10^{-3}. These results confirm that the proposed forward PINN framework can accurately reproduce a wide variety of stability behaviors, including both bistable and monostable configurations, as dictated by the underlying geometry.

3.2 Inverse design based on full energy curve

We then evaluate the inverse design capability of the proposed PINN framework by reconstructing CKO geometries that reproduce a prescribed target energy curve $\bar{U}(\bar{h})$. In contrast to the forward problem, where geometry is fixed and the deformation field is learned, the inverse design task treats both the structural parameters and the deformation field as trainable variables optimized simultaneously.

Fig. 5 presents the results for the four representative cases introduced in the forward prediction study. In each case, the target curve (red solid line), the PINN-predicted curve for the optimized geometry (blue dashed line), and the FEA validation results (black dots) are plotted over the normalized height domain. Across all cases, the three curves are in excellent agreement, indicating that the PINN-based inverse framework reliably identifies geometries whose mechanical response matches the desired target profile.

The optimized structural parameters for the inverse-designed geometries differ from those used to generate the target curves, reflecting the one-to-many mapping between energy landscape and geometry. Despite these differences in parameter values (Table S1), the resulting energy curves align almost perfectly with the targets, and FEA verification confirms that the designed structures are physically valid. Quantitatively, all cases achieve $R^2 > 0.99$ with respect to the target curves and $R^2 > 0.99$ in FEA validation, with RMSE values well below 1×10^{-3} (Table S2).

Fig. 6(A) shows the convergence behavior of the loss functions during training. For each of the four cases, the curves for the total loss, Euler–Lagrange loss, target matching loss, and physics constraint loss are plotted. The loss values decrease rapidly in the early stages and then enter a gradual refinement phase, eventually converging stably. In addition, Fig. 6(B) presents the evolution of the structural parameters (a, b, c, β) over the course of training. All four parameters exhibit stable convergence as training progresses, demonstrating the numerical stability and reliability of the proposed inverse design framework.

Overall, these results confirm that the proposed PINN-based inverse design framework can robustly reconstruct physically valid CKO geometries that reproduce prescribed energy curves with high accuracy, even when the optimal geometry differs from the original configuration.

3.3 Inverse design based on bistable energy programming

We further assess the flexibility of the proposed inverse design framework by considering an underdetermined setting in which only minimal energy information is specified. In this bistable programming task, the input consists of two zero-energy

Published on 07 November 2025. Downloaded by University of Minnesota - Twin Cities on 5/18/2026 4:06:03 AM.

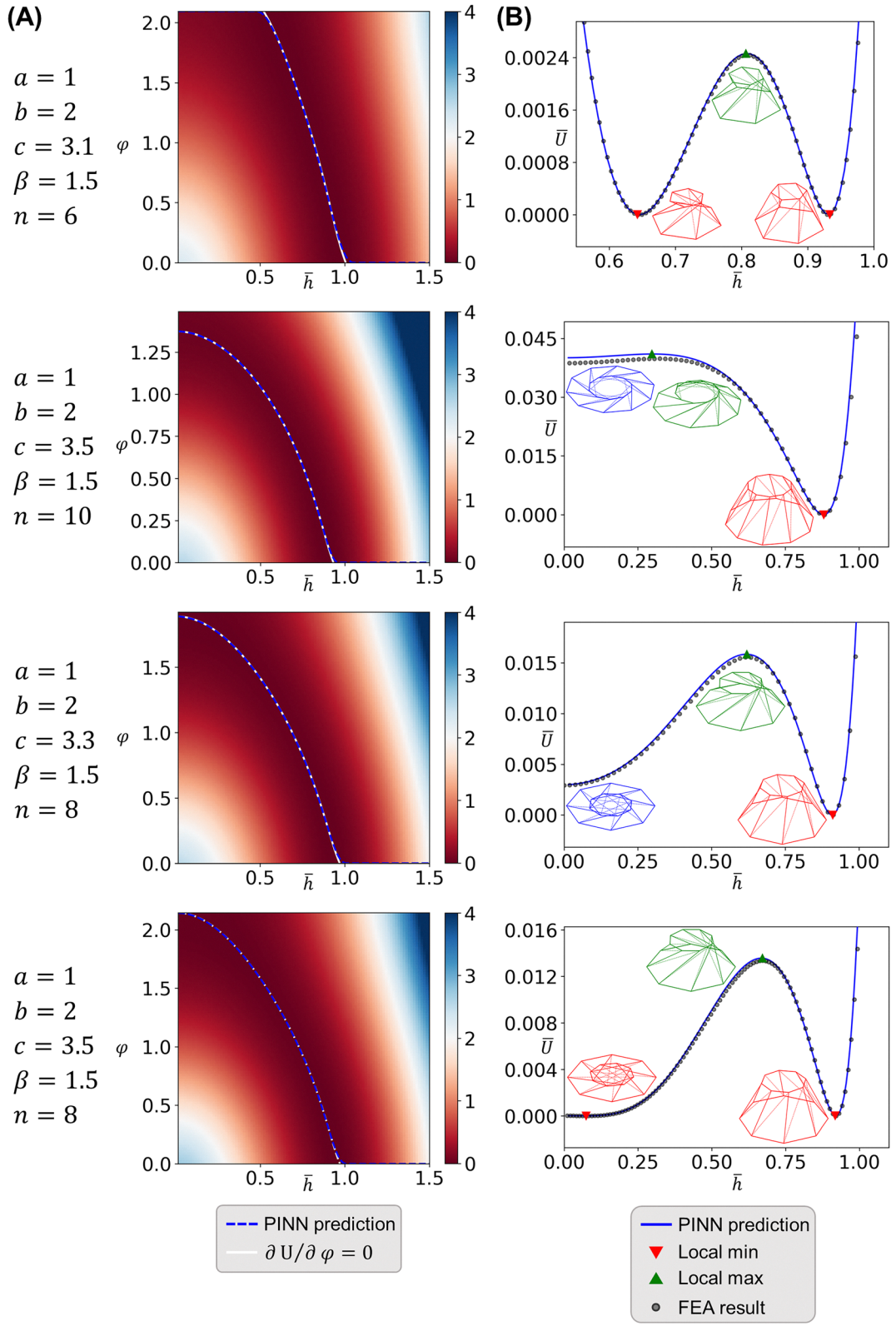


Fig. 4 Forward prediction of energy landscapes for diverse CKO geometries. (A) Energy fields $\bar{U}(\bar{h}, \varphi)$ for four representative cases with torque-free equilibrium paths (white solid lines: $\partial U/\partial\varphi = 0$) and PINN predictions (blue dashed lines). (B) Energy curves $\bar{U}(\bar{h})$ showing PINN predictions (blue solid lines) *versus* FEA results (black dots). Red triangles mark stable minima and green triangles indicate energy barriers, with corresponding 3D CKO structures embedded at these critical points. From top to bottom: Case 1 exhibits bistability with two zero-energy states; Case 2 shows monostability; Case 3 demonstrates asymmetric bistability with non-zero energy minimum (blue structure); Case 4 shows bistability with zero-energy states, one at $\bar{h} = 0$ (fully folded).

 This journal is © The Royal Society of Chemistry 2025

Published on 07 November 2025. Downloaded by University of Minnesota - Twin Cities on 5/18/2026 4:06:03 AM.

Table 1 Forward prediction accuracy for four representative CKO geometries. Coefficient of determination (R^2) and root mean square error (RMSE) values compare PINN predictions with FEA results for the normalized energy curves shown in Fig. 4(B)

Case	R^2 (FEA)	RMSE (FEA)
Case 1	0.991081	0.000216
Case 2	0.930294	0.004485
Case 3	0.997174	0.000256
Case 4	0.973334	0.001053

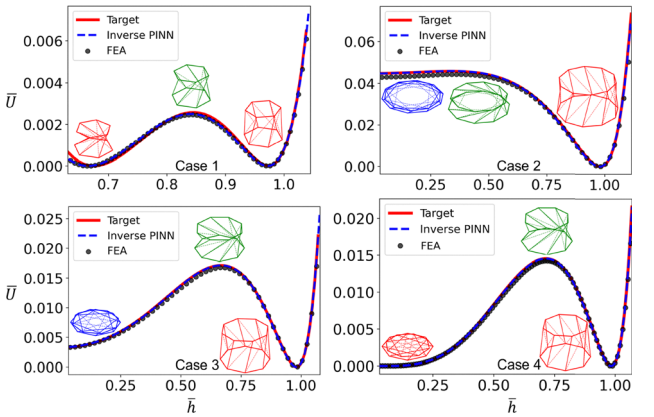


Fig. 5 Inverse design based on target energy curves. Comparison of normalized energy curves $\bar{U}(\bar{h})$ for four representative cases, showing target profiles (red solid lines), inverse-designed PINN predictions (blue dashed lines), and FEA validation results (black dots).

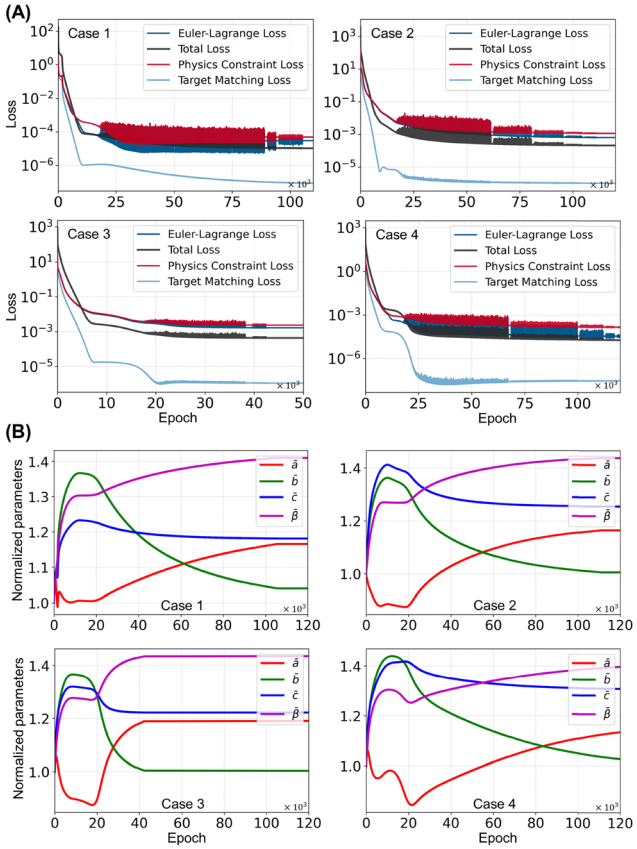


Fig. 6 Convergence behavior during inverse design training. (A) Evolution of loss components over training epochs for the four representative cases from Fig. 5. (B) Normalized structural parameters (a, b, c, β) plotted against training epochs, showing stable convergence trends.

stable heights and a target energy barrier, and the objective is to identify a physically feasible CKO geometry that satisfies these constraints.

To directly visualize the target stable positions and barrier height, we use the raw values of energy U and height h instead of normalized variables. Fig. 7(A) shows a representative case with a design domain $h \in [0, 4]$ mm, with stable points at h_1 = 0 mm and h_2 = 3.9 mm, and the target energy barrier set to 0.075 mJ. The inverse-designed energy curve (blue) matches the target profile, reproducing both zero-energy minima and the barrier height, and closely agrees with FEA validation (black dots) with R^2 = 0.9987 and RMSE = 0.000964.

Training convergence is illustrated in Fig. 7(B) and (C). All structural parameters (a, b, c, β) evolve smoothly toward stable values, and the total loss and its components decrease rapidly in the initial epochs and then gradually refine to reach stable values. Among these components, the barrier-loss curve exhibits a short non-monotonic fluctuation. This fluctuation arises because the barrier location is re-estimated at each epoch as the point of maximum energy within a predefined search region, and the identified maximum occasionally shifts to a neighboring sample on the discrete height grid as the rotation angle and geometric parameters evolve. The transient increase in barrier loss is also attributed to the ReLU-based curvature-sign constraint, where the gating term turns on or off when the local curvature sign temporarily changes. Despite this, the total objective continues to decrease throughout training, and all constraints converge stably, as shown in Section S4. These convergence plots confirm the numerical robustness of the inverse design framework for this challenging underdetermined problem. The comparison between the target and inverse-designed energy values at the stable points and barrier is summarized in Table S3 (SI Section S4), showing excellent agreement across all cases. The normalized absolute error remains below 1.4% in most configurations and does not exceed 2.3%, confirming the accuracy of the inverse design framework.

Fig. 7(D) and (E) further demonstrate the flexibility and programmability of the framework. In Fig. 7(D), five different cases are considered to evaluate performance under varying design targets. The first three cases fix the barrier height at 0.05 mJ and vary the second stable height: h_2 = 3.4, 3.9, and 4.4 mm, with design domains [0, 3.5], [0, 4], and [0, 4.5] mm, respectively. The remaining two fix h_2 = 3.9 mm and vary the barrier height: 0.05, 0.075, and 0.1 mJ. In all cases, the PINN-generated curves match the specified targets, and the corresponding 3D geometries at the barrier and second stable point are shown to the right of each plot.

In Fig. 7(E), we test whether the same energy targets (h_1 = 0 mm, h_2 = 3.9 mm, barrier = 0.075 mJ) can be achieved while varying the number of polygon units in the CKO geometry: n = 8, 9, 10. The resulting energy curves again satisfy the

Published on 07 November 2025. Downloaded by University of Minnesota - Twin Cities on 5/18/2026 4:06:03 AM.

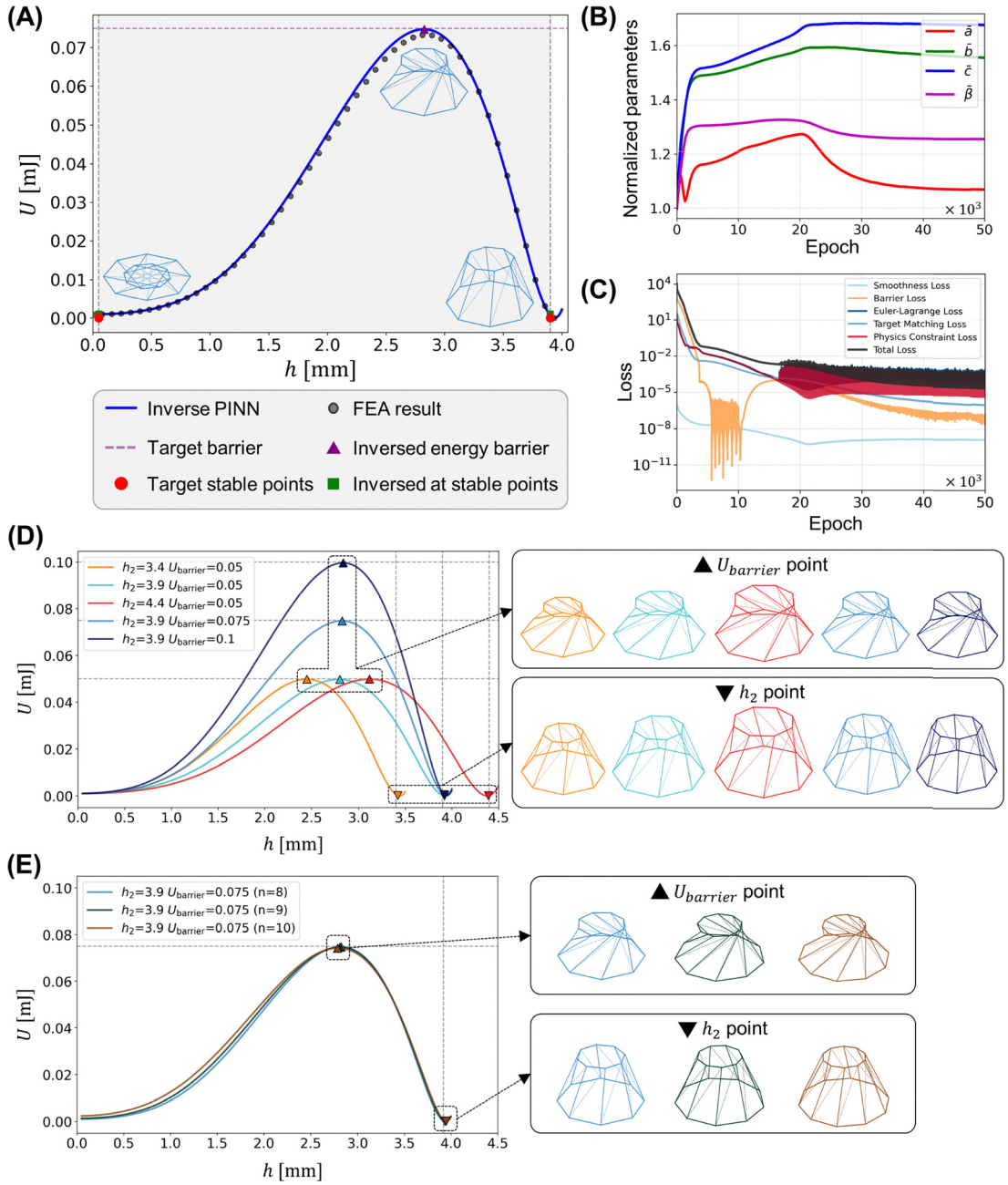


Fig. 7 Bistable energy programming with minimal input. (A) Representative case: h_1 = 0, h_2 = 3.9, $U_{barrier}$ = 0.075 mJ, domains [0,4]. (B) Evolution of normalized parameters (a, b, c, β) for case in (A). (C) Loss convergence for case in (A). (D) Five design variations with n = 8: three cases with h_2 = 3.4, 3.9, 4.4 (domains [0, 3.5], [0, 4], [0, 4.5]; $U_{barrier}$ = 0.05 mJ) and two cases with $U_{barrier}$ = 0.075, 0.1 mJ (domain [0, 4]; h_2 = 3.9). Right panels show 3D structures at barrier and second stable states. (E) Polygon number variation (n = 8, 9, 10) for identical targets: h_1 = 0, h_2 = 3.9, $U_{barrier}$ = 0.075 mJ, domain [0, 4]. Right panels show corresponding 3D structures.

targets, and FEA validation yields $R^2 > 0.99$ in all cases (Table S5). Full parameter values for all inverse-designed cases are provided in Table S4. Although the proposed framework demonstrates high flexibility, not all target barrier values can be physically realized. For a fixed pair of stable points h_1 and h_2, there exist upper and lower bounds dictated by the mechanics of the CKO unit, as discussed in SI Section S5.

These results demonstrate that the PINN framework can perform inverse design of bistable energy profiles under minimal specifications, enabling precise and programmable control over both stable positions and barrier heights, even in highly constrained design spaces.

3.4 Sequential deployment in multi-layer CKO structures *via* inverse design

We finally extend the inverse design framework to multi-layer CKO assemblies, aiming to achieve sequential deployment by prescribing distinct energy barriers for each layer. This capability

 This journal is © The Royal Society of Chemistry 2025

Published on 07 November 2025. Downloaded by University of Minnesota - Twin Cities on 5/18/2026 4:06:03 AM.

enables complex shape changes without active control, as each layer folds in a predetermined order dictated by its barrier height.

To illustrate the concept, we design three configurations: a cylinder, a cone, and a zigzag, each composed of three vertically stacked CKO units. To ensure sequential bottom-up folding, the energy barrier heights of the three layers are set to 0.05 (bottom), 0.08 (middle), and 0.11 (top), respectively. Fig. 8(A) shows the inverse-designed energy curves for all layers in each configuration, and the corresponding geometrical parameters are provided in Table S6. For each shape, the top and bottom radii of the units are fixed to achieve the intended overall profile.

The compression-induced energy evolution for all three configurations is presented in Fig. 8(B). In every case, folding initiates in the bottom layer, proceeds to the middle, and concludes with the top layer, matching the design targets. This staged collapse pattern is reproduced in FEA simulations, with strong agreement between simulated and designed energy landscapes. Movie S1 shows the dynamic deformation sequences. These results highlight that by shaping each layer's energy landscape and constraining its geometry, the framework can orchestrate multi-step deployments without active actuation. The same approach can be scaled to larger assemblies or adapted for temporal coordination in soft robotics and morphing structures.

To validate this multi-layer strategy, we fabricated and tested the zigzag configuration at a 16× scale based on the values in Table S6, to facilitate practical fabrication and testing. Fig. 9(A) shows the test setup for individual layers, alongside measured force–displacement and derived energy–displacement curves. As intended, the experimental barrier heights increase from Layer 1 to Layer 3, aligning with design targets (0.05, 0.08, 0.11 mJ). Quantitatively, the relative errors between normalized experimental values and PINN predictions are −7.93% for the middle layer and −16.16% for the top layer (Table 2). While the qualitative sequence is preserved, quantitative differences arise from physical effects absent in the truss-based model, such as crease rotation, polygon bending, and panel buckling, as observed in Movie S2. These mechanisms dissipate additional energy, leading to residual energy after each folding event rather than a full return to zero as predicted by the model. The cumulative effect appears in both individual layer tests and the complete assembly.

Fig. 9(B) presents the experimental validation of the complete three-layer zigzag assembly. The force–displacement curve reveals three distinct peaks corresponding to the sequential

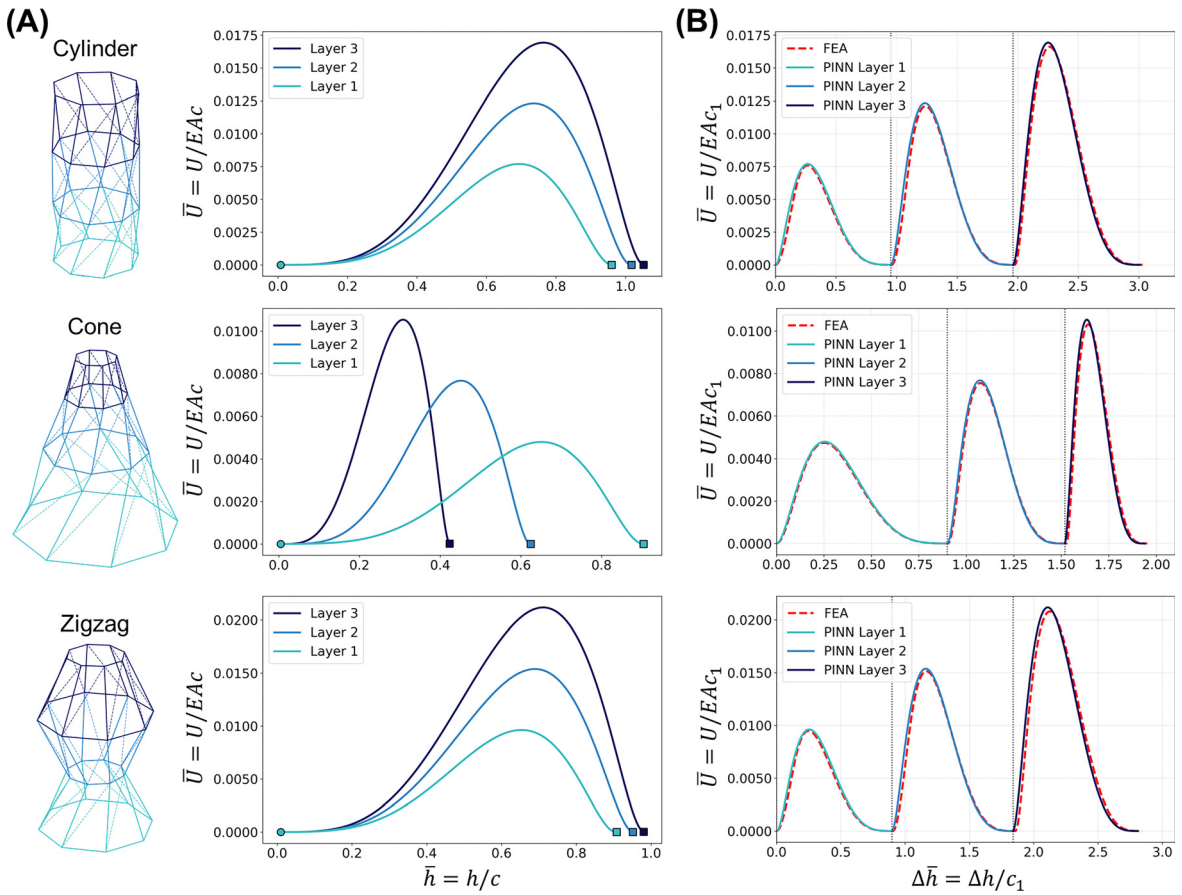


Fig. 8 Multi-layer CKO structures with programmable sequential deployment. (A) Three target geometries (cylinder, cone, zigzag) showing 3D structures and inverse-designed energy curves $\bar{U}(\bar{h})$ for individual layers. Energy barriers: 0.05 (bottom), 0.08 (middle), 0.11 (top). (B) Energy evolution $\bar{U}(\Delta\bar{h})$ during compression. Solid lines: PINN; dashed lines: FEA.

Published on 07 November 2025. Downloaded by University of Minnesota - Twin Cities on 5/18/2026 4:06:03 AM.

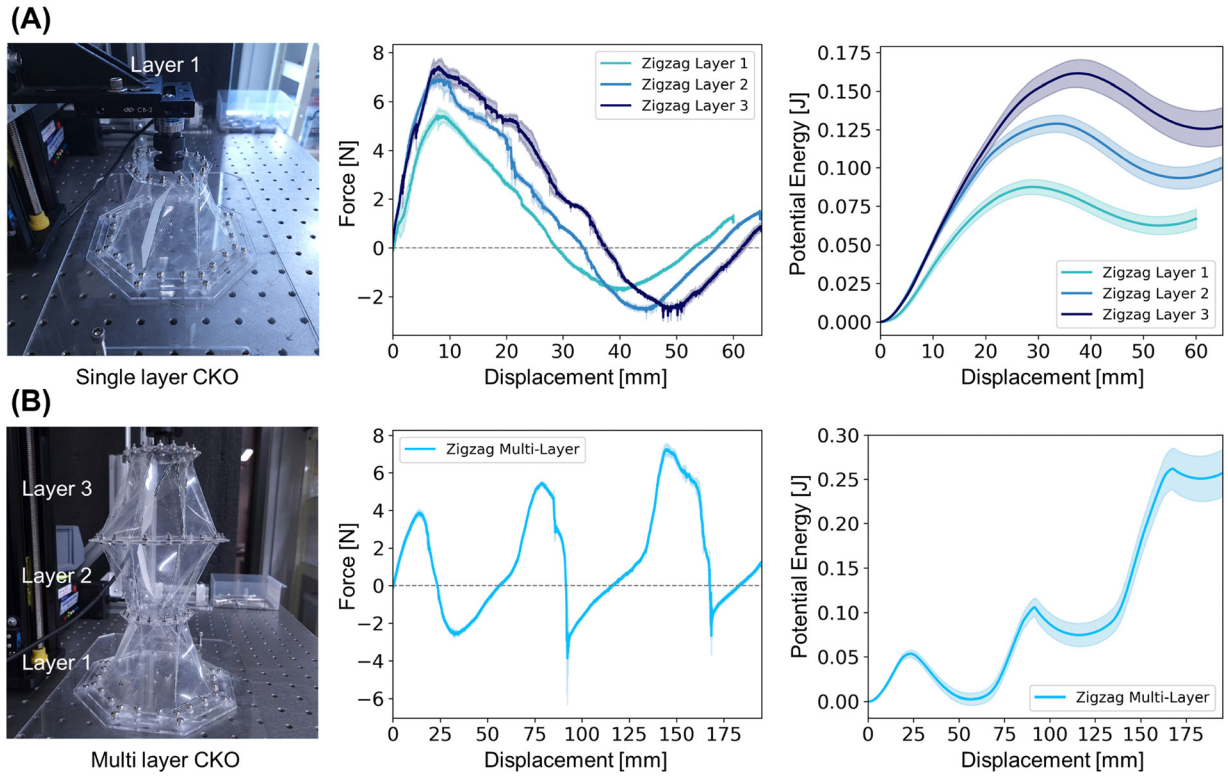


Fig. 9 Experimental validation of zigzag configuration. (A) Individual layer tests: layer 1 specimen photograph at initial state with force–displacement and energy–displacement curves for all three layers. (B) Three-layer assembly test: initial state photograph with force and energy response during compression showing three distinct peaks and energy accumulation.

Table 2 Comparison of PINN predictions and experimental energy barriers for each layer. Relative error is based on normalized experimental values; bottom layer is excluded as the normalization reference

Layer	PINN prediction	Experimental [mJ]	Normalized experimental	Relative error (%)
Bottom	0.05	0.0875	0.05	—
Middle	0.08	0.1289	0.0737	−7.93
Top	0.11	0.1614	0.0922	−16.16

folding of each layer, with peak magnitudes increasing from bottom to top. The accompanying energy profile confirms the progressive accumulation effect observed in individual tests, wherein the energy does not return to zero between folding events but stabilizes at elevated levels. This behavior preserves the intended sequential deployment in the designed order (*i.e.*, from the bottom to the top layer), with progressively increasing energy barriers, thereby validating our design strategy. Movie S2 provides a dynamic visualization of the compression experiments, capturing the deformation of each individual layer and the complete multi-layer assembly. In the multi-layer test, the sequential folding sequence is clearly evident, with each layer folding in the predetermined order.

Overall, these experiments confirm that despite simplified modeling assumptions, the inverse-designed energy landscapes translate effectively to real structures, enabling robust sequential deployment with programmable barrier hierarchies.

4. Discussion

This study establishes a comprehensive PINN framework for designing CKO with programmable mechanical behavior. The framework unifies forward prediction and inverse design without pre-collected training datasets, addresses longstanding challenges in modeling multistable origami mechanics, and clarifies how computational assumptions map to observed physical responses. Together with FEA and benchtop experiments, the results show that energy landscapes, such as stable states, barrier heights, and deployment paths, can be shaped directly through geometry.

The physics-informed strategy differs fundamentally from conventional data-driven approaches: instead of learning from labeled examples, it embeds mechanical equilibrium and energy principles into the loss so that predictions satisfy torque-free conditions and variational consistency by construction. These inductive biases improve sample efficiency and interpretability and enable extrapolation to regions of design space not explicitly seen during optimization. Extending the formulation from cylindrical to conical geometries further broadens the attainable design space by allowing different top/bottom radii and, consequently, richer kinematic couplings between axial motion and twist.

A key contribution is bistable energy programming from minimal specifications. Traditional inverse design often requires full target curves or many constraint points; our method needs

 This journal is © The Royal Society of Chemistry 2025

Published on 07 November 2025. Downloaded by University of Minnesota - Twin Cities on 5/18/2026 4:06:03 AM.

only two stable heights and a desired barrier value, infers the barrier location automatically, and recovers the full energy landscape consistent with physics. This reduces the cognitive and computational burden on the designer while retaining control over deployment forces and states. Importantly, the same idea scales to hierarchical assemblies: by prescribing layer-wise barrier magnitudes, we achieve predictable, bottom-up folding in multi-layer CKO systems without active feedback or complex controls. From a practical standpoint, the results suggest a simple design rule: to ensure robust sequencing under real-world dissipation, maintain a sequence margin between adjacent barrier heights that exceeds expected experimental deviations (*e.g.*, the top layer in our tests showed the largest underestimation). Such margins help preserve the intended order even when non-ideal effects accumulate.

Experiments highlight both strengths and limitations of the current model. The truss-based abstraction captures deployment order and relative barrier magnitudes, but quantitative differences arise because real prototypes exhibit deformation modes omitted in the model: rotational stiffness at creases, bending of polygonal frames, panel buckling, frictional/contact effects, and associated dissipation. These mechanisms explain the observed energy accumulation between folding events, whereas the idealized model predicts a return to zero. To improve quantitative agreement and extend physical fidelity, future refinements of the energy formulation should incorporate additional deformation modes such as rotational (crease) stiffness, local buckling, hinge torsion, and frame/panel bending and twisting. These effects can be integrated while retaining computational tractability and full compatibility with the current PINN structure. Two complementary routes are attractive: (i) a gray-box extension that augments the analytic energy with a low-dimensional learned discrepancy term regularized by physics, and (ii) multi-fidelity training, in which the PINN is calibrated to FEA but nudged toward experiment with sparse measurements. Both approaches can reduce bias while preserving the interpretability and stability benefits of physics-informed learning.

Beyond model form, there are methodological considerations. Inverse design is one-to-many: multiple geometries can realize the same energy curve. Our framework currently selects one feasible solution; in practice, designers may prefer solutions that optimize secondary criteria such as manufacturability (feature sizes, angle ranges), robustness to tolerances, or material economy. These can be incorporated as soft penalties or additional objectives (*e.g.*, minimum-norm parameters, sensitivity-aware regularization that penalizes large barrier gradients with respect to a, b, c, β, n). Likewise, initialization and nonconvexity can affect convergence; multi-start strategies, curriculum schedules for loss weights, and parameter nondimensionalization are pragmatic safeguards that we found to promote stable optimization. Finally, while our study focuses on quasi-static behavior, many applications involve rate effects and dynamics; extending the framework to time-dependent PINNs with damping and inertia, or to hybrid simulators that combine our energy model with reduced-order dynamics, would enable trajectory-level design (*e.g.*, speed of deployment, impact mitigation).

The broader implications extend to mechanical metamaterials and deployable structures in which energy landscape design is the primary lever. Aerospace systems can benefit from passive, fail-safe sequencing in booms, reflectors, and morphing skins; soft robotics can exploit programmed barriers to realize snap-through actuation, mechanical memory, or force-limited behaviors; impact-mitigation devices can tune threshold energies for staged absorption. Because the formulation is fast, differentiable, and interpretable, it lends itself to design-automation workflows and interactive tools that allow rapid exploration of alternatives and real-time prediction. Looking forward, coupling the framework with uncertainty quantification (*e.g.*, Bayesian PINNs or ensemble surrogates) would enable reliability-aware designs that maintain sequencing under fabrication variability. Incorporating fabrication-aware constraints (minimum crease length, perforation patterns, allowable thickness) and contact/friction models, as well as interlayer coupling effects such as residual stress accumulation, axial or rotational mismatch between neighboring layers, and contact activation during compression, will further improve predictive accuracy and better align the model with experimental observations. Accounting for these effects within the total potential formulation would allow the framework to capture the redistribution of strain energy and barrier shifts observed in multilayer tests more faithfully, ultimately reducing the remaining gap between PINN predictions and real multilayer folding behaviors.

5. Conclusions

This work presents a PINN framework that addresses both forward prediction and inverse design of CKO without relying on pre-collected training data. By embedding mechanical equilibrium and energy principles directly into the learning objective, the framework produces interpretable, physically consistent designs across a broad range of geometries and operating conditions.

First, we develop a data-free learning approach that predicts and designs complex energy landscapes governed by multistability. In forward prediction, the framework accurately captures monostable and bistable behaviors across diverse geometries, closely matching FEA with $R^2 \geq 0.93$ and small RMSE values on the order of 10^{-3}. In inverse design against full target curves, agreement is even tighter, with $R^2 \geq 0.99$ relative to both target profiles and FEA validations and RMSE below 10^{-3}. Together, these results demonstrate high fidelity while eliminating the need for labeled datasets.

Second, we introduce an inverse-design methodology that programs energy barriers from minimal specifications, namely two stable heights and a target barrier value, while autonomously inferring the barrier location and recovering a full, physically feasible energy profile. The method generalizes across stable-state spacing, barrier magnitude, and polygon count n, and identifies feasible solutions even in underdetermined settings. We also delineate that realizable barrier heights are bounded for a fixed pair of stable states, clarifying practical design limits.

Published on 07 November 2025. Downloaded by University of Minnesota - Twin Cities on 5/18/2026 4:06:03 AM.

Third, we extend the approach to hierarchical, multi-layer assemblies and realize programmable sequential deployment by prescribing layer-wise barrier magnitudes. FEA confirms bottom-to-top folding in the intended order for cylinder, cone, and zigzag configurations. Benchtop experiments on a three-layer zigzag prototype (scaled 16×) reproduce the designed sequencing and relative barrier hierarchy. The relative errors of the normalized experimental barriers are −7.93% and −16.16% for the middle and top layers, respectively. While quantitative differences arise from deformation modes not included in the truss model (crease torsion, frame bending, panel buckling, friction/contact), the qualitative deployment order and barrier progression are preserved, supporting real-world robustness.

The established framework provides a foundation for fabrication-aware, physics-consistent design of origami-inspired metamaterials. Immediate directions include enriching the energy functional with hinge torsion and frame/panel bending, incorporating contact/friction models, and adopting gray-box or multi-fidelity training to reduce model-experiment bias while retaining interpretability. Integrating uncertainty quantification and sensitivity-aware penalties can improve robustness to tolerances and preserve sequencing margins under variability. Extending to time-dependent settings (damping/inertia) will enable trajectory-level design for rate-sensitive applications.

In summary, the proposed PINN framework turns geometric parameters into programmable energy landscapes, ranging from single units to hierarchical assemblies, and bridges theory and practice in origami engineering. Its efficiency, generality, and experimental viability point to broad impact across deployable aerospace structures, soft-robotic actuators, impact-mitigation devices, and biomedical mechanisms where precise, adaptable mechanical responses are essential.

At the same time, the present formulation relies on a simplified truss abstraction that omits certain physical effects such as facet bending, crease torsion, and frictional contact. These simplifications may lead to minor quantitative deviations from experiments, particularly in highly deformed or multilayer configurations. In addition, geometric constraints and stiffness asymmetry can cause slight under- or over-estimation of energy barriers, defining practical limits on achievable designs. Incorporating these effects into the total potential formulation and extending the framework to nonlinear and contact-rich regimes will further enhance predictive accuracy and broaden the applicability of physics-informed learning to complex origami metamaterials.

Author contributions

Sukheon Kang: conceptualization, data curation, formal analysis, investigation, methodology, software, validation, visualization, writing – original draft, writing – review & editing. Youngkwon Kim: formal analysis, investigation, methodology, validation, writing – review & editing. Jinkyu Yang: formal analysis, methodology, writing – review & editing. Seunghwa Ryu: conceptualization, funding acquisition, methodology, project administration, supervision, writing – review & editing.

Conflicts of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All code and data used in this study are openly available on Zenodo, accessible at: **https://doi.org/10.5281/zenodo.16991596**.

Supplementary information is available. See DOI: **https://doi.org/10.1039/d5mh01607j**.

Acknowledgements

This work was supported by the National Research Foundation of Korea (NRF) (No. RS-2023-00222166) and the InnoCORE program (No. N10250154), both funded by the Ministry of Science and ICT (MSIT), as well as by a grant from the Ministry of Food and Drug Safety (No. RS-2023-00215667). Y. K. and J. Y. acknowledge the support from the Air Force Office of Scientific Research (FA2386-24-1-4051) and the National Research Foundation of Korea (2023R1A2C2003705 and No. 2022H1D3A2A03096579).

Notes and references

1 M. J. Islam, B. Bao and F. Peng, *Appl. Mater. Today*, 2025, **44**, 102715.
2 M. Meloni, J. Cai, Q. Zhang, D. S.-H. Lee, M. Li, R. Ma and J. Feng, *Adv. Sci.*, 2021, **8**, 2000636.
3 Z. Zeng, T. Li, H. Dong, L. Yang, T. Liu and X. Chen, *Front. Mech. Eng.*, 2025, **20**, 12.
4 Z. Song, J. F. Zhu, X. Wang, R. Zhang, P. Min, W. Cao and C. W. Qiu, *Nat. Commun.*, 2024, **15**, 3181.
5 W. Xue, B. Jian, L. Jin, R. Wang and Q. Ge, *Adv. Mater. Technol.*, 2025, e00278.
6 T. Kim, E. A. Dunn, M. Chen and R. L. Truby, *Adv. Mater.*, 2025, e01290.
7 Y. Liu, W. Shi, P. Chen, Y. Yu, D. Zhang and D. Wang, *Front. Mech. Eng.*, 2023, **18**, 54.
8 J. O'Neil, M. Salviato and J. Yang, *Compos. Struct.*, 2023, **304**, 116376.
9 Z. Chen, X. Wei, L. Yang and J. Xiong, *J. Mech. Phys. Solids*, 2024, **188**, 105672.
10 S. Yao, H. Zhou, T. Gao, F. Chen, Z. Wang and K. Liu, *Int. J. Mech. Sci.*, 2025, **290**, 110114.
11 J. Gao and G. J. McShane, *Int. J. Solids Struct*, 2025, 113585.
12 H. Wang, D. Zhao, Y. Jin, M. Wang, T. Mukhopadhyay and Z. You, *Appl. Mater. Today*, 2020, **20**, 100715.
13 Z. Zhai, L. Wu and H. Jiang, *Appl. Phys. Rev*, 2021, **8**, 041319.

 This journal is © The Royal Society of Chemistry 2025

Published on 07 November 2025. Downloaded by University of Minnesota - Twin Cities on 5/18/2026 4:06:03 AM.

14 B. Kresling and J. F. Abel, in Proc. 6th Int. Conf. Comput. Shell Spatial Struct., International Association for Shell and Spatial Structures, Ithaca, NY, 2008, vol. 11, pp. 12–32.
15 S. M. Alipour and J. Arghavani, *Mech. Adv. Mater. Struct.*, 2024, **31**, 11755–11770.
16 M. Yang, J. Defillion, F. Scarpa and M. Schenk, *Acta Mech. Solida Sin.*, 2023, **36**, 514–530.
17 H. Sharma, A. Chaudhary and S. H. Upadhyay, *Thin-Walled Struct.*, 2023, **190**, 110980.
18 S. Mora, N. M. Pugno and D. Misseroni, *Extreme Mech. Lett.*, 2025, **77**, 102314.
19 S. Wu, T. Zhao, Y. Zhu and G. H. Paulino, *Proc. Natl. Acad. Sci. U. S. A.*, 2024, **121**, e2322625121.
20 Z. Li, Z. Wang and W. D. Wang, *ACS Appl. Mater. Interfaces*, 2024, **16**, 7850–7859.
21 B. Tian, Z. Yan, Q. Li, X. Hu and T. Tan, *Mater. Horiz.*, 2023, **10**, 2398–2411.
22 N. Gao, X. Liu and J. H. Wu, *Compos. Struct.*, 2025, 119523.
23 W. Qiang, H. Feng, T. Zhou, G. Lu and X. Zhang, *Int. J. Mech. Sci.*, 2024, **282**, 109682.
24 J. O'Neil, M. Salviato, E. Kim and J. Yang, *Int. J. Solids Struct*, 2025, **309**, 113199.
25 J. Zhang, L. Zhang and C. Wang, *Int. J. Mech. Sci.*, 2022, **227**, 107470.
26 L. Lu, X. Dang, F. Feng, P. Lv and H. Duan, *Proc. R. Soc. A*, 2022, **478**, 20210712.
27 S. Zang, D. Misseroni, T. Zhao and G. H. Paulino, *J. Mech. Phys. Solids*, 2024, **188**, 105630.
28 H. Yasuda, T. Tachi, M. Lee and J. Yang, *Nat. Commun.*, 2017, **8**, 962.
29 C. Jiang, D. Wang, P. Cheng, L. Qiu and C. Li, *Mech. Mach. Theory*, 2024, **203**, 105796.
30 Y. Yin, Y. Hu, Y. Yu, Y. Zhang, C. Liu, W. Sun and B. Li, *Phys. Rev. Appl.*, 2025, **23**, 034070.
31 D. D. Lim, A. Ibarra, J. Lee, J. Jung, W. Choi and G. X. Gu, *Sci. Adv.*, 2025, **11**, eads3499.
32 J. Park, J. Noh, J. Shin, G. X. Gu and J. Rho, *Adv. Funct. Mater.*, 2025, **35**, 2412901.
33 S. Kang, H. Song, H. S. Kang, B. S. Bae and S. Ryu, *Mater. Des.*, 2024, **247**, 113377.
34 H. Yasuda, K. Yamaguchi, Y. Miyazawa, R. Wiebe, J. R. Raney and J. Yang, *Commun. Phys.*, 2020, **3**, 168.
35 Y. Zhu and E. T. Filipov, *Sci. Rep.*, 2022, **12**, 19277.
36 S. Han, C. Li, Q. Han and X. Yao, *Extreme Mech. Lett.*, 2025, **74**, 102276.
37 D. Zhang, A. K. Qin, Y. Chen and G. Lu, *Eng. Struct.*, 2023, **278**, 115497.
38 M. Raissi, P. Perdikaris and G. E. Karniadakis, *J. Comput. Phys.*, 2019, **378**, 686–707.
39 H. Jeong, C. Batuwatta-Gamage, J. Bai, Y. M. Xie, C. Rathnayaka, Y. Zhou and Y. Gu, *Comput. Methods Appl. Mech. Eng.*, 2023, **417**, 116401.
40 J. Yin, Z. Wen, S. Li, Y. Zhang and H. Wang, *Comput. Methods Appl. Mech. Eng.*, 2024, **426**, 117004.
41 J. He, T. Chen, G. Li, W. Chu and T. Xu, *Phys. Lett. A*, 2025, **533**, 130213.
42 C. P. Batuwatta-Gamage, C. M. Rathnayaka, H. C. P. Karunasena, H. Jeong, M. A. Karim and Y. T. Gu, *Int. J. Mech. Sci.*, 2024, **275**, 109267.
43 X. Chen, Y. Wang, Q. Zeng, X. Ren and Y. Li, *Ocean Eng.*, 2025, **319**, 120260.
44 C. X. Liu, X. Wang, W. Liu, Y. F. Yang, G. L. Yu and Z. Liu, *Int. J. Mech. Sci.*, 2024, **269**, 109080.
45 G. Wan, Y. Cai, Y. Liu, C. Jin, D. Wang, S. Huang, N. Hu, J. X. J. Zhang and Z. Chen, *Extreme Mech. Lett.*, 2021, **42**, 101065.
46 Y. Zhang, M. Tichem and F. van Keulen, *Extreme Mech. Lett.*, 2022, **51**, 101553.
47 J. A. Faber, J. P. Udani, K. S. Riley, A. R. Studart and A. F. Arrieta, *Adv. Sci.*, 2020, **7**, 2001955.
48 G. Librandi, E. Tubaldi and K. Bertoldi, *Nat. Commun.*, 2021, **12**, 3454.
49 H. Sharma and S. H. Upadhyay, *Mech. Based Des. Struct. Mach.*, 2023, **51**, 113–137.
50 Y. Miyazawa, C. W. Chen, R. Chaunsali, T. S. Gormley, G. Yin, G. Theocharis and J. Yang, *Commun. Mater.*, 2022, **3**, 62.
51 Y. Miyazawa, H. Yasuda and J. Yang, *Acta Mech. Sin.*, 2023, **39**, 723169.